AI Civilization — Visual Overhaul v4

From 'detailed but unreadable' to 'genuinely cool'. A showcase-grade visual plan: legibility first, then a crafted palette, real lighting, an isometric world, and juice. All procedural Pygame — no sprite art, no asset pipeline. ~1 month at 45–60 min/day.

The honest diagnosis

The simulation is complete (v5.0, 294 tests, all six arcs + minds-at-the-pivots) and the renderer has plenty of detail — but the current look fails the showcase test for four reasons: **noise over signal** (the event feed is trust-delta spam; major events drown), **flat rendering** (top-down circles and icons read as a diagram, not a world), **undisciplined colour** (many saturated hues at similar values turn overlapping settlements into mush), and **weak lighting** (night is a dark tint, not darkness with light in it). The fix is not more objects — it is rendering craft: hierarchy, palette, light, projection, motion.

Design north star

A stranger watching only the map should follow the story beat by beat — and want to keep watching. Reference feel: the readable-clean of Mini Metro, the mood of a lantern-lit strategy map, the depth of a 2.5D isometric world. Simple shapes, crafted like a designed game — not decorated like a tech demo.

Ground rules (unchanged)

- Renderer stays READ-ONLY; byte-identical seeded sim runs; AST boundary green; all 294 tests pass every slice.
- All visuals procedural (shapes, polygons, gradients) — no image assets, no sprite art.
- One slice at a time, committed clean; visual slices are verified by EYES plus the suite, never the suite alone.
- Performance budget: smooth at grid 24 / 30 agents (~10ms frame); cache aggressively, per-frame cost stays flat.

Week 1 — Legibility & Signal (fix what's broken first)

Iso-projection on an unreadable scene is lipstick on noise. Week 1 makes the current look READ, and fixes two live bugs.

V4.1 — HUD + Realms bugfix. The bottom HUD draws overlapping garbled text; the REALMS panel shows '(none)' while STATE shows kingdoms 2 / empires 1. Fix HUD layout (proper spacing at any zoom) and wire the realm scoreboard to actually read kingdoms/empires with settlement counts in realm colours.

V4.2 — Event tiers + the story banner. MAJOR events (battles, coronations, uprisings, secessions, empire formed/fragmented, era advances, faith founded, ruler deaths) always show, larger and bolder; MINOR noise (per-agent trust deltas, routine trades) is aggregated to one line per turn or suppressed. A prominent top-of-map BANNER announces each major event in plain words ('KING Borin's host was REPELLED at S0A2') — the single feature that makes the map self-explanatory without the side panel.

V4.3 — Territory clarity. Overlapping settlement circles currently blur into a blob. Distinct outlined regions: stronger boundary stroke, lower fill opacity, realm-coloured edges; labels collision-nudged off buildings. Night floor brightness raised and weather particle density/opacity reduced so atmosphere never obscures the story.

End state: the current renderer, readable — a viewer follows the story from map + banner alone.

Week 2 — Palette & Light (the crafted look)

V4.4 — Palette discipline. Pick 4–5 base hues and push every element toward them; desaturate the commons; reserve saturation for what matters (rulers, banners, battles, the banner strip). One PALETTE pass over terrain, buildings, agents, territory and UI so the whole frame reads as one designed image. This is taste work — expect 2–3 tuning evenings with screenshots side by side.

V4.5 — Real light sources. Replace the flat night tint with LIGHT AS A SYSTEM: windows, torches, hearths and the castle brazier become point lights casting warm radial gradients onto the ground (cached gradient surfaces, additive blend). A town at night becomes a cluster of warm pools in blue darkness; dawn dissolves the pools. Battles at night are lit by the clash flashes. This is the single biggest MOOD upgrade available per hour spent.

End state: screenshots that look colour-graded and lit — a lantern-lit world, not a tinted map.

Week 3 — The Isometric World (the big jump)

The one genuinely structural slice: tilt the world into 2.5D. Same procedural shapes, new projection — buildings gain HEIGHT, castles TOWER, terrain gains depth. This is what moves the look from 'diagram' to 'game'. It touches every draw call, the camera, town plans and the cinematics — treat it like the camera slice: one shared projection function, built carefully.

V4.6 — The projection core. One pure world_to_screen_iso((x, y, z), cam) used by EVERY draw call (diamond-tile ground projection, z for height). Draw order becomes painter's-algorithm by projected depth (back-to-front). Camera pan/zoom and the minimap carry over through the same transform. Terrain rebaked as diamond tiles with elevation shading.

V4.7 — Buildings with height. Town-plan structures re-drawn as simple 3D forms: wall faces (lit side / shade side per the global light direction), roof planes, the castle keep and towers genuinely TALLER than houses, the palisade a ring of posts with height. Agents become upright figures standing ON the ground plane, shadows anchored at their feet.

V4.8 — Iso polish + cinematics re-seat. War cinematics, speech bubbles, banners, crowns, trails, weather and the day/night light re-seated in the projection (bubbles float above heads; rain falls TOWARD the ground plane; territory drawn as ground overlay under buildings). Verify every prior feature reads correctly in iso before calling it done.

End state: a tilted, deep, lit world — the screenshot people stop scrolling for.

Week 4 — Juice & the Showcase Cut

V4.9 — Juice. Small motion that makes it feel alive: subtle screen shake on battle clashes, dust/particle burst when a building rises or falls, a gentle zoom-punch when the story banner fires, 1px squash-and-stretch on agent steps, banner cloth ripple. Each effect subtle; together they read as production value.

V4.10 — Showcase mode. A --showcase flag: runs the most cinematic staged scenario, auto-glides the camera to major events (banner fires -> camera eases there), clean title overlay at start, HUD minimal. Built for screen recording — this is the mode the devlog footage comes from.

End state: press record, get footage that looks like a real game trailer.

Cadence

Week	Theme	Slices	The visible win
1	Legibility & signal	V4.1–V4.3	Story readable from map + banner alone; bugs gone
2	Palette & light	V4.4–V4.5	Colour-graded frames; lantern-lit nights
3	Isometric world	V4.6–V4.8	2.5D depth — buildings tower, world tilts
4	Juice & showcase	V4.9–V4.10	Trailer-grade footage on demand

Risk notes (stated up front)

• V4.6 (iso projection) is the heavy slice — a genuine rework touching every draw call. If it overruns, steal days from Week 4; juice is shrinkable, the projection is not.

• Palette and lighting are TASTE slices: budget explicit tuning evenings; screenshots before/after are the test, not the suite.

• Iso will surface layout bugs (overlaps, z-fights) the way the season slice surfaced the cache test — expect a fix-forward day after V4.7.

• Hard scope line: NO sprite art, no external assets, no 3D engine. The ceiling is crafted procedural 2.5D — which is enough to look genuinely cool.

What this is for

The simulation already earned its showcase — a world that produced Rex the Grasping, Kade the Liberator, and an uprising that recorded why it rose. This plan makes the picture worthy of the story: readable in week 1, crafted in week 2, dimensional in week 3, filmable in week 4. Then record the devlog from --showcase and publish.